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ORIGINAL PAPER

Anaerobic Co-digestion Mill Wastewater—Activated Sludge Effect of Aerobic Pretreatment on the Performance of OMW Anaerobic Digestion

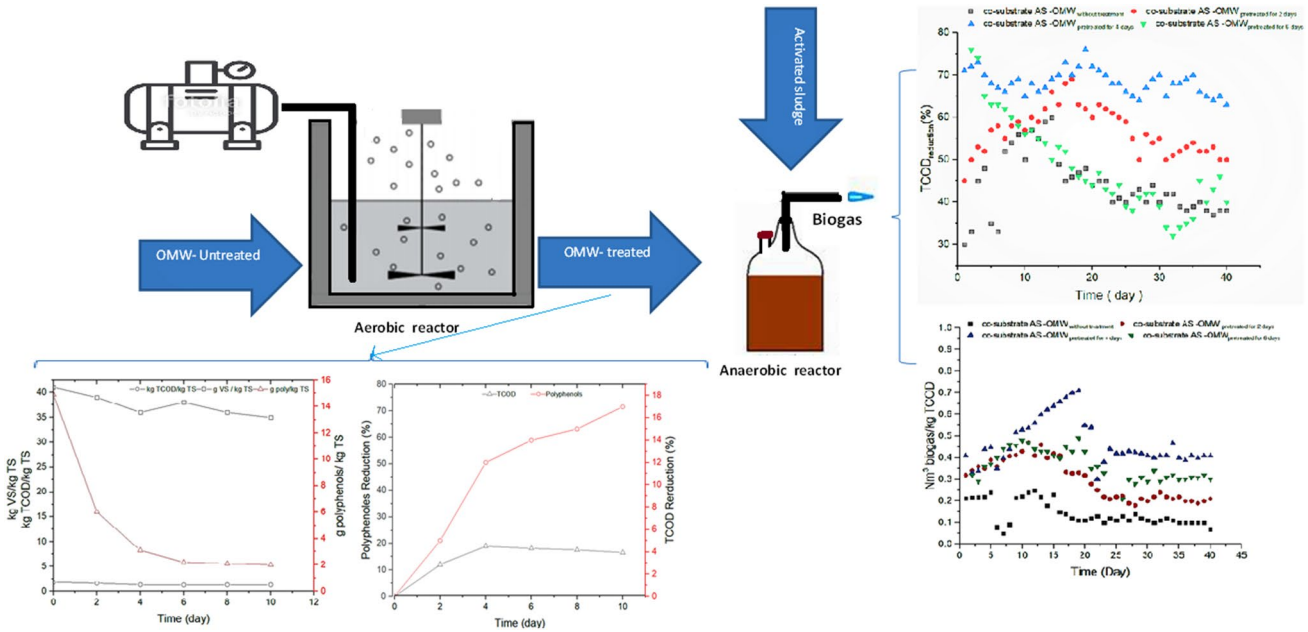
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Abstract

Aerobic treatment of olive mill wastewater (OMW) is a promising technique for biogas recovery using anaerobic digestion, in the case of OMW it is possible to reduce polyphenols and TCOD concentration which undesirable by microorganism metabolism. OMW was subjected to the aeration process over different periods to detect the removal efficiency of polyphenols and TCOD. During this study, anaerobic co-digestion of OMW without pretreatment and activated sludge (AS) was inhibited due to the presence of a high inhibitors concentration. While anaerobic co-digestion of aerated OMW–AS were achieved TCOD removal 56.34%, 70.32%, 52.81% and polyphenols removal 42.69%, 60.45%, 63.51% when the OMW aerated for 2, 4 and 6 days respectively. The highest biogas production 0.473 Nm^3/kg VS was achieved by OMW aerated for 4 days in conjunction with 68.39% CH4, 55.85% VS removal, 10.91 g/l of TCOD and 4.23 g $CaCO_3$/l.

Graphic Abstract



Keywords The aerobic process · Anaerobic co-digestion · Olive mill wastewater · Activated sludge · Polyphenols removal · VS removal · COD reduction

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Statement of Novelty

This study provides new data about the potential of pretreatment of olive mill wastewater co-digestion with activated sludge. Pretreatment conducted by the aerobic process and detected the best period to applied aerobic process to achieve higher COD and polyphenols reduction. This technique maximized the biogas production conjunction with high VS, TCOD and polyphenols removal efficiency.

Introduction

Among the possible renewable energy sources, biowaste from activated sludge (AS) is a promising resource [1]. An attractive process solution for cancelling accumulation of active sludge is the anaerobic digestion (AD) process [2–5]. Therefore, AD of the AS with higher volatile solids (VS) content has been considered to be an alternative to biomass digestion without competition to food production [6, 7].

Thus, VS content higher than 4% in comparison to the conventional AD processes from other sources which have TS 0.5–3%, could operate by using existing AD infrastructure at wastewater plants [2–4]. Moreover, the short residence time for complete digestion of AS from 16–46 days were reported when total solid (TS) with removal rate 32% of VS increased from 1.79 to 15.67% [8].

The AD inhibitors AS were believed to be directly related to the decreased capacity of functional microorganisms. Microbial communities may also differ due to the accumulation of volatile Fatty Acids (VFA) combined with high, thus leads to alternation in metabolic pathways and decrease AD performance. On the other hand, various suggestions were presented to avoid inhibition through co-digestion technology [3, 6, 7, 9, 10]. In practical, using AD to co-digest AS with other biowaste materials to improve biogas production has been proposed and implemented at the wastewater plant [2–5], that can be achieved by using existing infrastructure at the wastewater plant without extra capital cost.

The world production of olive oil in 2017 about 2.945 million tons and according to provisional figures by the International Olive Council (IOC), the Global olive oil production is increased to rise by 28% for 2018. The by-products of Olive oil mills such as olive mill wastewaters (OMW), could be considered as suitable feedstock for the AD process. Actually, those by-products of Olive oil mills pose a serious risk, where the organic compounds measured as 40 g/l to 220 g/l [11, 12]. However, OMW was subjected to many treatment studies in terms of physical and chemical treatment [13–17], one of the satisfactory solutions to dispose of these organic pollutants is the AD process [18, 19].

However, the AD of OMW suffered low pH value, high content of phenolic, and ammonium nitrogen lack, which lead to inhibition. Currently, to overcome these limitations, pretreatments for OMW have been reported such as aerobic process and evaporation in storage ponds. In order to encourage bacterial growth, aeration as a pretreatment process had been applied with different technologies. That approves digestion process efficiency and high polyphenols rates [20]. The Polyphenol molecular structure is concerned as one of the main toxicity reasons. When the Hydroxyl Number Group (HNG) decreases the inhibitory effect is increased, HNG is associated with polyphenols concentration [21].

Many studies showed that OMW could be improved with low dilution by drying without adding chemical substances, also could be subject to co-digestion process with AS which contains a high level of ammonium nitrogen. Co-digestion of AS with OMW provides the necessary nutrients and buffer capacity also decrease potential inhibitors of the AD process [16–22].

The biogas production and methane content are based on an improved composition of the substrate to be ready for AD, since each one of the substrates complements other deficiency. Biogas production is significantly increased when co-substrate is used [23–28].

This work study reports on the use of the pretreatment technique of OMW by aerobic digestion. Also, presents an investigation of the pretreated OMW effect on the performance of codigestion process with AS, biogas and methane production. Previous tests of aerobic before AD process were performed to determine the suitable time of aeration to achieve maximum COD and polyphenols reduction. Aerated OMW with ratio 50% volume/volume to AS were carried out by the Biomethane Potential (BMP). This technique was implemented by other authors through AD and aerobic processes, achieving high efficiency of the digestion process in terms of biogas production and eliminating organic load [16, 17, 29].

Methodology—Materials and Methods

Feedstock

OMW was extracted from a two-phase oil extraction that belongs to the olive harvest seasons of 2017. OMW was collected in 20 l Jerrycan. AS was obtained from secondary sludge of a wastewater treatment plant WWTP of Irbid city (Actual load in the range of 8635.1–9000 m^3/d and 8276.9 m^3/d of hydraulic load outside), Jordan. OMW and AS were

stored at 4 °C until use. According to UNI 5667-13/2000, all the samples used in the BMP tests were prepared. Co-substrates of pretreated and non-treated OMW and AS were homogenized using a manual shaking (2 times /day for 1 min) and were prepared on a volume (v/v) basis. Raw materials were analyzed of both physico-chemical characteristics and proximate analysis through TGA [30]. The characteristics include the humidity U, total solid TS, viscosity solid VS, ASH and fixed Carbon F.C, thus analysis was optioned according to the standard methods as described Alrawashdeh et al. [5]. The raw material characteristics are summarized in Table 1.

Experimental Setup

A manual batch aerobic reactor was prepared which consists of a vessel of 3 l realized in Broro-silicate glass and has the injection of air is fixed at the bottom of the vessel using an air compressor in which the flow rate of air is 0.5 l/l substrate/min. The temperature of the substrate was room atmosphere (25–28 °C).

The BMP tests were prepared in closed vessels of 1 l (Borosilicate glass) which are conducted with two necks, one to extract and introduce the feedstock, and the other to collect the generated biogas during the process. Vessels were reproducibility of the tests in order to reliable results.

The biogas volume generated during the experiment is measured using the biogas measurement method as described in [31]. The generated biogas was investigated and analyzed by GC as described Alrawashdeh et al. [5]. The vessels operated in semi conditions under mesophilic conditions (35 ± 2 °C) using water bath surrounding vessels. The vessel content of which was homogenized using manual shaking, allowing for complete mixing. In order to avoid conditions of pressurized, the generated biogas in excess was vented through extraction neck using siring.

Experimental Procedure

A set of aerobic tests were carried out to investigate the time of aeration that achieves maximum reduction of COD and polyphenols of the OMW substrates. Afterwards, AD tests were performed using pretreated and non-treated OMW as co-substrate with AS substrate without adding any inoculum, given that anaerobic bacteria already present in the AS substrate.

The samples of OMW substrate were prepared in separated experiment tests, i.e., one test from 2 to 10 days. In each test, the aerobic reactor was filled up to 1 l (50% of reactor volume). Before the AD test all, the pretreated OMW substrates were sorted at 4 °C.

To identify the effects of the aerobic process as a pretreatment technique to achieve higher anaerobic digestion efficiency, a set of parameter measurements was performed during the process. Thus parameters, including TCOD, TS and polyphenols reduction and pH value "hydrogen ion concentration". The samples of pretreated OMW were analyzed to identify the best period of aeration.

Next, all the AD vessels were filled up to 20% of their volume (200 ml) with different co-substrate of OMW-AS prepared as recommended in similar studies [5]. A set of the mixture was utilized 50% v/v of AS -pretreated OMW (for different periods 2–10 days), in addition, 50% v/v of AS–OMW (non-treated), in order to investigate the aeration effects on biogas production and TCOD reduction.

The AD tests were operating in semi-continuous mode, the volume of OMW was added daily into the vessel and the same quantity was extracted from the vessel, therefore the flow rate of 10 ml of OMW/ day. The tests were performed for 40 days, biogas, methane and degradation level were obtained. The maximum degradation and Hydraulic Residence Time (HRT) are defined based on the time that bacterial metabolism takes place. The 1st HRT was the adaptation period of in which the VS concentration was increased due to co-substrate replaced by OMW during introduce and extraction substrate. While the 2nd HRT was stability period that presents the steady state degradation, therefore the effluent analysis was showed decreases of TCOD reduction, biogas production was stabilized and the pH values were constant.

Produced Biogas was investigated using water replacement method [31], the compositions of biogas were analyzed and detected for each test. Both samples of influent and effluent were analyzed to detect the U, TS, VS, ASH and F.C, in addition, the values TCOD reduction and pH. Table 2 shows the characteristics and parameters of the operating of pretreated OMW and co-substrate which fed to AD tests. Proximate analysis was performed using Thermographic Analysis (TGA), the TCOD was detected using Nano colour kits and spectrophotometer (Macherey–Nagel).

Produced biogas was sampled using gas-tight syringes (250 µl) and collected via rubber stopper of the small neck of the vessel. Samples of biogas were analyzed in duplicate by a gas chromatograph (490 micro GC, Agilent Technologies, SC, CA, USA. GC was used to detect the quantity of the (CH_4, CO_2, CO, C_2, O_2 and N_2). GC consists of a thermal

Table 1 Composition of olive mill wastewater (OMW)—activated sludge (AS)

Substrate	U (%)	TS (%)	VS (%)	Ash (%)	F.C (%)	pH
OMW	88.3 ± 0.5	11.7 ± 2.2	4.8 ± 1.2	4.39 ± 3.5	2.51 ± 2.9	4.4 ± 0.3
AS	95.57 ± 0.31	4.43 ± 4.1	2.45 ± 1.5	1.98 ± 1.9	0 ± 1.3	6.8 ± 0.5

Table 2 Composition of pretreated and non-treated olive mill wastewater (OMW)—activated sludge (AS)

Co-substrate 50% v/v	U (%)	TS (%)	VS (%)	Ash (5)	F.C (%)	pH
AS+OMW non-treated	91.93 ± 1.41	8.07 ± 2.9	3.63 ± 0.6	3.19 ± 1.3	1.26 ± 2.6	4.9 ± 1.1
AS+OMW Pretreated—2 days	92.39 ± 1.23	7.61 ± 1.7	3.23 ± 1.9	3.19 ± 2.4	1.19 ± 3.3	5.8 ± 1.04
AS+OMW Pretreated—4 days	93.11 ± 0.09	6.89 ± 4.2	3.19 ± 2.1	3.13 ± 1.0	0.58 ± 1.5	6.3 ± 0.9
AS+OMW Pretreated—6 days	93.14 ± 0.51	6.86 ± 1.03	2.93 ± 0.31	3.18 ± 3.01	0.76 ± 0.04	6.6 ± 1.21

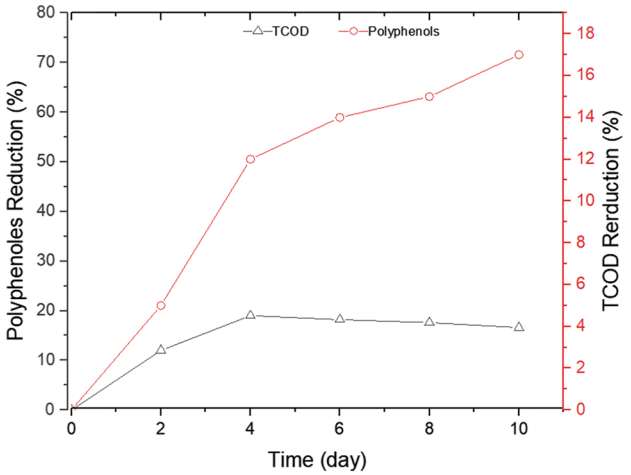


Fig. 1 The result of the pre-treatment process by aerobic digestion in the term of percentage of TCOD and polyphenols reduction

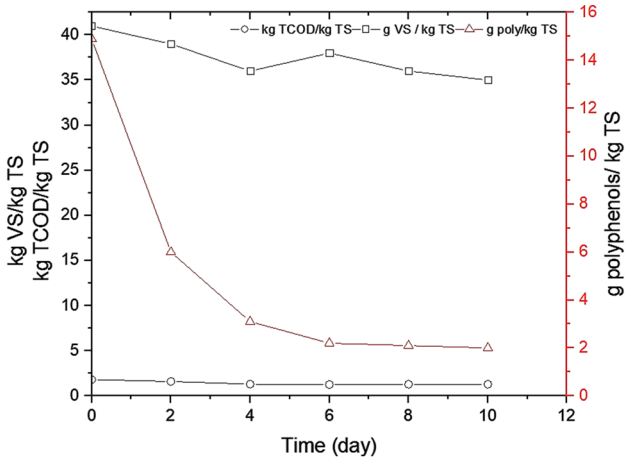


Fig. 2 Result of the pre-treatment process by aerobic digestion in the term of concentration of VS, TCOD and polyphenols reduction per Kg of TS

detector and capillary column (column injector and detector were operated at 80 °C, 100 °C and 180 °C). The carrier gas is He and Ar with a flow rate of 10 ml/min.

Results and Discussion

The main parameter measurements during the aerobic process were obtained as shown in Figs. 1 and 2. By tracing the graph it can be observed that the effect of increasing aeration period was reflected on the TCOD reduction, polyphenols concentration and VS reduction. There was a decreasing trend in the concentration of polyphenols. In the fourth day it was reduced to 4.1 g/kg TS, but the reduction after 6 days it remained in the range of 0.1–0.2 g/kg TS and the downward trend continued over the following days.

An intense increase in the TCOD reduction was also noticed during the process, where the range of reduction was 1–17% from the 1st to the 10th day of the aeration period, and the greatest reduction of the TCOD was observed from the 1st day to 4th day with 12%. As shown Fig. 2 the concentration of TCOD, polyphenols and VS were measured as the function of TS, A slight decreasing in VS concentration was noticed and after the 4th day, there was a small portion of decrement as a result of the stability of aeration process. It should be noted that during aerobic degradation there were decreasing in TS as a result of evaporation in conjunction with aerobic degradation.

Give the polyphenols results the maximum reduction 50% was achieved in 4 days and from 4th day to the 6th day was achieved 1% while from 6th day to 8th day only 10%. So, increase the time of aeration after the 4 days was not feasible in term of polyphenols reduction. The same results were noticed in the case of TCOD, where the TCOD reduction can be described as a stable period compared to pre-day 4.

Investigating the results, it can be shown that the maximum reduction rate of polyphenols was found to be around 50%. This reduction was achieved in the first 4 days. From the 4th to 6th day and from 6th to 8th days, a reduction of 11% was achieved, whereas a reduction of 10% was achieved from the 8th to 10th days (in total 70% for 10 of aeration). From these results, it can be shown that the reduction polyphenols verse the aeration time isn't directly proportional and the cutoff period, for best reduction results is 4 days.

Over the aeration time, an increase in the pH value of the OMW substance was observed from 4.4 to 5.2 without modifying their pH value. The maximum pH value was achieved in the 4th day, also C/N ratio was detected during the process which its value was in the range of 22 to 36. Similar studies were conducted to detect the effect of

aerobic treatment of olive mill wastewater, thus confirmed with our result in this study. El Hajjouji et al. [29] reported that aeration treatment of OMW could increase the efficiency of polyphenols reduction, where the maximum reduction was achieved about 94% in junction with pH about 5.99 and 5.99 of C/N. Essays made by Gonzalez and Cuadros [16] showed that the pretreatment was used with providing air in the flow rate of 0.65 l/l reactor, this study achieved maximum polyphenols degradation for 7 days, which were the same trend in our treatment for 4 days. In another study Michailides et al. [20], the researchers used aeration of OMW with continuous recirculation under one day of HRT, this technique reduced 50% of the concentration of phenolic, according to them increase the aeration time does not achieve a higher reduction in polyphenols concentration.

Given the results of pretreatment tests of OMW, four anaerobic digestion tests (with duplicates to investigate the accuracy of the result) were prepared using co-substrate of OMW and AS. One vessel filled up to 20% of its volume with 50% of non-treated OMW and 50% AS, other vessels filled with the same configuration, but instead, non-treated OMW in each test used pretreated OMW for 2 days, 4 days and 6 days. However, the tests didn't include pretreated OMW sample because as mentioned before, after the 6 days of aeration time it had reached stability in the case of TCOD degradation and lowest reduction in the case of polyphenols. Table 3 shows the characteristics and the configuration of co-substrate were used in anaerobic digestion.

The anaerobic tests were conducted for 40 days with 25 days of HRT, where the organic loading ORL flow rate was 10 ml OMW/ day for all tests, ORL of OMW was the same as a configuration of the OMW used in each test.

The biogas yields of all tests shown in Fig. 3, where the biogas production considered as a function of TCOD removed from the co-substrate. Overall, the anaerobic co-digestion process, the first 25 days were the acclimation period, then reached steady state for the co-digestion of pretreated OMW and AS. While co-digestion of untreated OMW–AS was suffered from inhibition after 15 days as a result of high polyphenols and TCOD concentration and

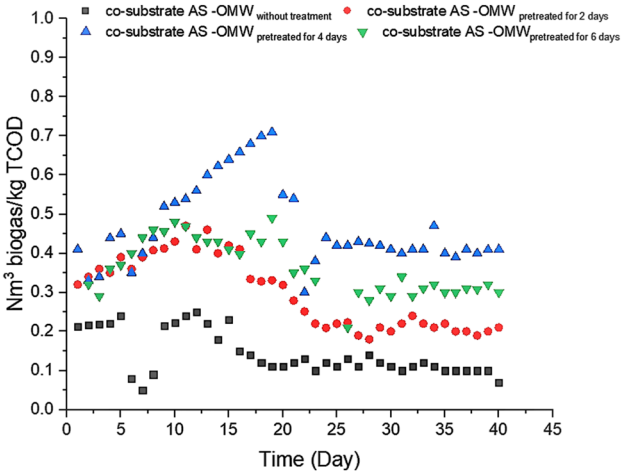


Fig. 3 Biogas production during anaerobic co-digestion of WAS-AS

low value of pH. It should be indicating that even if OMW is added to AS (rich in anaerobic bacteria and has high pH value), the digestion unviable.

Co-digestion of OMW aerated for 2 days showed low biogas production at the initial period then after 20 days was inhibited due to buffer accumulation as a result of organic load and polyphenols concentration. Similarly, as illustrated in Fig. 3, co-digestion of OMW aerated for 6 days was trended to produce biogas at a high rate during HRT period then dramatically decreased. While co-digestion of OMW aerated for 4 days showed the highest biogas yield that related to a lower concentration of polyphenols and TCOD and high pH value.

Tables 3 and 4 specify the results and characteristic of co-substrate and digested, as illustrated high dissolved efficiencies of TCOD removal (70.33%) achieved by co-digestion of OMW aerated for 4 days and AS, while the non-aerated OMW achieved the lowest efficiency of TCOD removal.

Table 3 illustrates the monitoring pH value for all digested. During the untreated OMW–AS test period, the pH was not stable 4.9–6.1 in which unfavourable for bacterial, while during aerated OMW and AS digestion pH was approximately stable. Figure 4 shows the

Table 3 Biogas production of co-substrate and characteristics of co- substrates and digested

Co-substrate 50% v/v	Biogas (Nm^3/kg_{TCOD})	$TCOD_{Co\text{-}sub}$ (g/l)	$TCOD_{digested}$ (g/l)	Alkalinity (g_{CaCO3} /l)	pH
AS (100%)	0.123 ± 0.32	23 ± 04	29.12 ± 0.42	7.22 ± 0.1	6.52 ± 1.04
OMW (100%)	0.101 ± 0.25	45.0 ± 0.52	38 ± 0.31	8.8 ± 1.02	5.8 ± 0.32
AS+OMW non-treated	0.132 ± 0.41	44.7 ± 0.58	36.03 ± 0.03	8.1 ± 1.01	6.1 ± 2.1
AS+OMW pretreated—2 days	0.291 ± 0.23	39.9 ± 0.41	19.66 ± 0.02	6.65 ± 1.21	6.91 ± 0.93
AS+OMW pretreated—4 days	0.473 ± 0.03	27.03 ± 0.36	10.91 ± 0.04	4.23 ± 1.65	7.24 ± 0.5
AS+OMW pretreated—6 days	0.329 ± 0.02	26.9 ± 0.5	12.73 ± 0.02	5.42 ± 1.08	8.04 ± 0.14

Table 4 characteristics of co-digestion OMW pretreated—AS and OMW non-treated—AS

Co-substrate 50% v/v	Biogas (Nm^3/Kg_{VS})	CH_4 (%)	VS-removal (%)	COD-removal (%)	Polyphenols removal (%)
AS (100%)	0.13	50.46	27.22	46.1	30.25
OMW (100%)	0.095	23.2	21.54	32.87	18.65
AS+OMW non-treated	0.107	25.8	22.36	44.9	25.03
AS+OMW pretreated—2 days	0.286	45.16	35.69	56.34	42.69
AS+OMW pretreated—4 days	0.44	68.39	55.82	70.32	60.45
AS+OMW pretreated—6 days	0.352	60.58	48.45	52.81	63.51

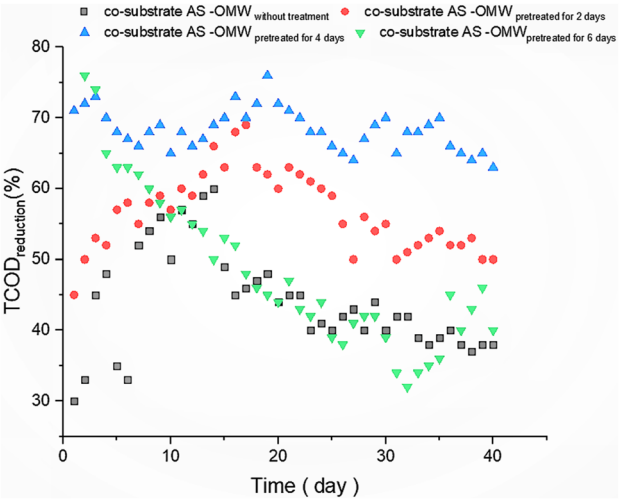


Fig. 4 TCOD reduction during anaerobic digestion

concentration of TCOD during the anaerobic co-digestion of WAS-AS which reflects the effect of aerobic digestion as pretreatment.

VS reduction was used as assessing the performance of the anaerobic digestion process, during the process VS degraded then converted to biogas. Among co-digestion OMW aerated for 4 days and AS showed the highest VS reduction 0f 55%, which agrees with 70.32% of TCOD removal and 60.45 of polyphenols removal. Also, this result showed that as the organic load increased, it presents the lowest biogas production and the lowest capability of degradation.

However, the result of anaerobic tests showed that co-digestion with untreated OMW lead to inhibit and even destabilize the process of anaerobic, this inhibition associated with polyphenols which lead to the toxicity of substrate which clearly appeared in the performance of digestion process. For this reason OMW substrate is required, that agrees with El-Hajjouji et al. [29]; Olivier et al. [10]; Belaid et al. [32]; Oz and Uzun [33] and Alrawashdeh [31].

Conclusion

Olive mill wastewater has a high concentration of polyphenols and Alkalinity balance lack; thus aerobic treatment of OMW could be a promising perspective on anaerobic digestion AD. This technique can achieve the highest removal of pollutants. The result of AD showed that the highest polyphenols removal efficiency 60.45% was achieved by OMW aerated for 4 days, where the results show, as increase the aeration time, the polyphenols reduction increasing.

Also, the results show that the optimum aeration period is 4 days for anaerobic digestion process, which it was achieved the highest biogas production 0.44 Nm^3/kg VS with quality 68.39% of CH4 in the conjunction of 55.82% and TCOD removal 70.32%.

The biogas production of untreated OMW with AS was inhibited (0.132 Nm^3/kg_{COD}, 8.1 g_{CaCO3}/l of Alkalinity concentration and 36.03 g/l of TCOD) due to high polyphenols and pollutants continent and low pH value which were led to the lowest VS degradation. Using co-digestion of pretreated OMW and AS improve the pH value, C/N ratio, an organic load required from methanogenic bacteria. However, the organic loading rate of pretreated OMW should be low in order to avoid conditions of overloading and buffer accumulation which leads to inhibit the AD process.

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